Capstone 2 Final Report
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Introduction

Today's retailers possess formidable tools that were not available just a few decades ago. In the past, retailers relied solely on their own observations and intuitions to discern market trends and draw conclusions for business decisions. However, with the advent of modern data science methods, online retailers can harness the vast dataset they already collect about their customers to develop predictive models and conduct in-depth analysis. This enables them to make informed decisions across various business areas, including marketing and product development.

Retailers who fail to leverage available data in this manner find themselves at a competitive disadvantage and risk losing market share to their competitors.

Problem Statement

Our client is an online apparel retailer seeking to segment its customer base using session-level clickstream and product-attribute data (the "e-shop clothing 2008" dataset, which includes session IDs, click sequences, product categories, colors, price information, and user country). The objective is to uncover distinct customer segments characterized by browsing behaviors, product preferences, and demographics in order to enable targeted marketing campaigns and personalized outreach. Success will be defined by generating clusters with a silhouette score ≥ 0.5, ensuring each segment is both well-separated and actionable for future campaign design.

Data Source

For this project, I utilized a dataset sourced from the UC Irvine Machine Learning Repository. This dataset comprises clickstream data pertaining to an online retailer based in Poland that specializes in maternity attire. The dataset spans a period of five months during the year 2008.

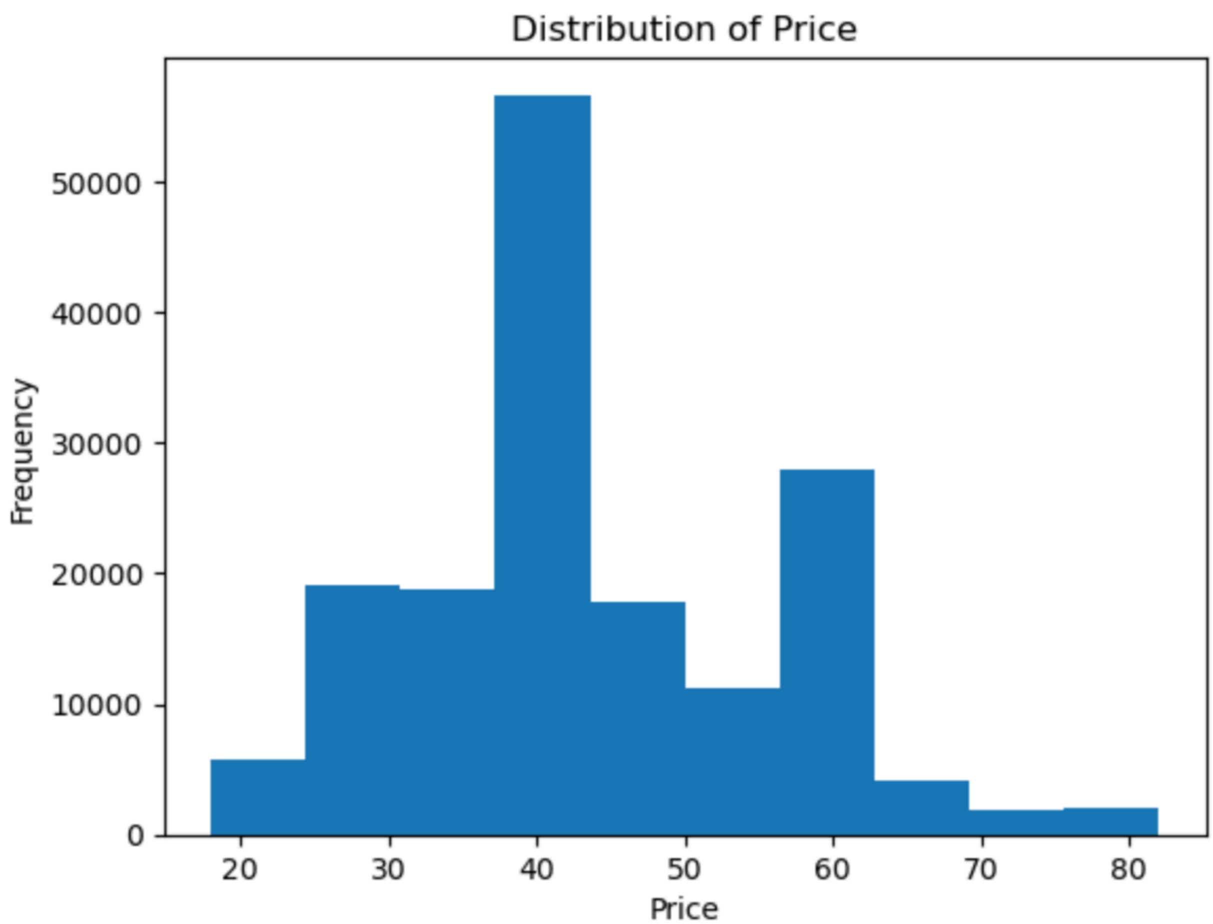
https://archive.ics.uci.edu/dataset/553/clickstream+data+for+online+shopping

Data Wrangling and Cleaning

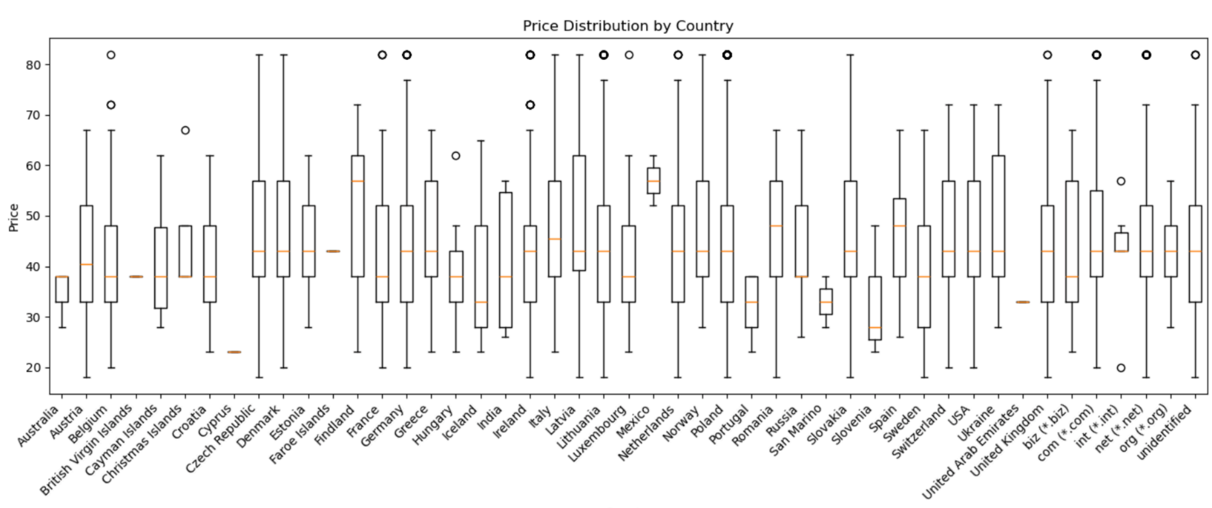
During the data preparation phase, I identified no missing values or duplicates in the dataset. To enhance usability, I created a date column derived from the individual year, month, and day columns. Upon plotting by country, it became evident that approximately 80% of purchases originated from Poland, the website's home country. An additional 10% originate from the Czech Republic, while the remaining 10% come from various other regions. Furthermore, I converted the price column from an integer to a float to facilitate future calculations.

Exploratory Data Analysis

To begin, I converted the appropriate columns to categories as many of the columns were formatted as integers. Subsequently, I reviewed the summary statistics for the price column and the value counts for the various categories. This analysis revealed that the price distribution of the items is skewed towards approximately $40, with the second largest price range centered around $60.



During the outlier exploration, I also analyzed the price distribution across countries, as presented in the following chart.



Based on the data exploration, several relationships have emerged. It is evident that displaying an item on the first page significantly increases its purchasing likelihood. Additionally, items positioned in the top middle and top left corners of a page exhibit a higher purchasing probability.

Furthermore, a distinct trend is observed among items categorized as olive, red, blue, pink, and black, which exhibit an average price that surpasses the overall mean price of $43.80.

It is also noteworthy that there are 1,923 items with an outlier price of $82.00.

Considering that the business in our case study is situated in Poland, it is unsurprising that the majority of purchases originate from Polish customers. Given the limited number of purchases from various countries, it is not advisable to explore subgroups based on nationality.

Model Selection

During the modeling phase, I evaluated K-Means, Agglomerative Clustering, and DBSCAN models to identify the most suitable clustering method for determining effective clusters for future targeting marketing campaigns.

The results indicated that K-Means yielded an optimal elbow between 3 and 4, but the silhouette score at 2 and 3 remained low, suggesting weak clusters. K-Means resulted in large, diffuse groups that lacked a clear association with behavioral patterns.

Hierarchical clustering, on the other hand, did not provide a natural "cut" height to create cohesive clusters, as evidenced by the dendrogram. This model, like K-Means, yielded weak clusters.

Finally, DBSCAN resulted in clusters with sizes ranging from approximately 100 to 44,000. The moderate silhouette score indicated no strong clustering. The visualization of the data revealed micro-clustering, with a large cluster containing all data points that did not adhere to a micro cluster.

Algorithm	Parameters	n_clusters	n_noise	Silhouette Score	Calinski-Harabasz Score
MiniBatchKMeans	n_clusters=2, batch_size=2000	2	0	0.186	
AgglomerativeClustering	n_clusters=2, linkage=ward	2	0	0.192	
DBSCAN	eps=1.2820, min_samples=10	96	100	0.406	4234.882

Conclusions and Next Steps

The exploration and modeling of the data provided useful insights around which prices are most popular and where our current customer base is located. It also provided insights into which page location is most likely to lead to a purchase and which colors are most popular.

Further experimentation and modeling are necessary to identify clusters that can be utilized to achieve our initial objectives of segmenting data for marketing purposes. This can be accomplished through experimentation with various models and hyperparameter optimization.